

Network Packet Capture and Protocol Analysis Report

Tool Used: Wireshark

Interface: eth0

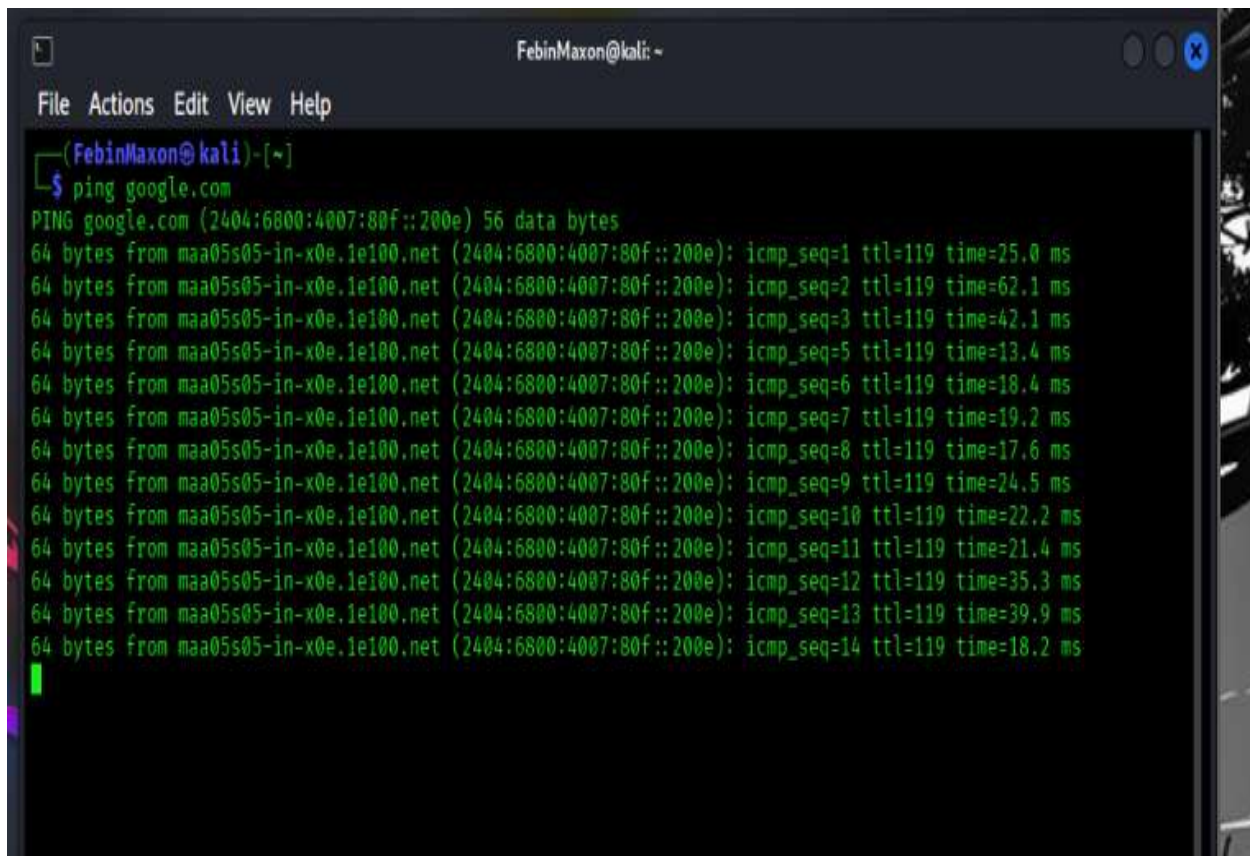
1. Objective

The objective of this task was to capture live network packets using Wireshark and identify different network protocols and traffic types generated during normal network activity.

2. Methodology

- Wireshark was installed and launched on the system.
- Packet capture was started on the active network interface (`eth0`).
- Network traffic was generated by executing the command:
- Additional traffic was generated by background applications and browsing activity.
- Packet capture was stopped after approximately one minute.
- Protocol-based filters such as `icmp` and `dns` were applied to analyze the captured traffic.
- The captured packets were saved as a `.pcap` file.

`ping google.com`



```
FebinMaxon@kali: ~  
File Actions Edit View Help  
(FebinMaxon@kali)-[~]  
$ ping google.com  
PING google.com (2404:6800:4007:80f::200e) 56 data bytes  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=1 ttl=119 time=25.0 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=2 ttl=119 time=62.1 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=3 ttl=119 time=42.1 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=5 ttl=119 time=13.4 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=6 ttl=119 time=18.4 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=7 ttl=119 time=19.2 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=8 ttl=119 time=17.6 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=9 ttl=119 time=24.5 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=10 ttl=119 time=22.2 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=11 ttl=119 time=21.4 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=12 ttl=119 time=35.3 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=13 ttl=119 time=39.9 ms  
64 bytes from maa05s05-in-x0e.1e100.net (2404:6800:4007:80f::200e): icmp_seq=14 ttl=119 time=18.2 ms
```

3. Protocols Identified

At least **three different protocols** were identified in the capture:

1. ICMP (Internet Control Message Protocol)

- Used for network diagnostics and reachability testing.
- Generated by the `ping google.com` command.

Observations:

- ICMP Echo Request packets sent from local IP `192.168.0.181`
- ICMP Echo Reply packets received from Google server IP `142.250.77.142`

- TTL values around 119
- Round Trip Time (RTT) ranged between approximately 6 ms – 9 ms
- No packet loss observed (0%)

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=0/0], ttl=64 (request id 1)
2	0.000111	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=0/0], ttl=64 (reply id 1)
3	0.000340	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=4/1924, ttl=64 (request id 12)]
4	0.000490	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=4/1924, ttl=64 (reply id 12)]
5	0.000693	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=8/1200, ttl=64 (request id 52)]
6	0.000745	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=8/1200, ttl=64 (reply id 52)]
7	0.000950	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=12/536, ttl=64 (request id 125)]
8	0.001002	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=12/536, ttl=64 (reply id 125)]
9	0.001208	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=16/2700, ttl=64 (request id 100)]
10	0.001260	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=16/2700, ttl=64 (reply id 100)]
11	0.001465	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=20/2040, ttl=64 (request id 187)]
12	0.001517	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=20/2040, ttl=64 (reply id 187)]
13	0.001722	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=24/2880, ttl=64 (request id 288)]
14	0.001774	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=24/2880, ttl=64 (reply id 288)]
15	0.001979	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=28/2304, ttl=64 (request id 100)]
16	0.002031	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=28/2304, ttl=64 (reply id 100)]
17	0.002236	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=32/2560, ttl=64 (request id 170)]
18	0.002288	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=32/2560, ttl=64 (reply id 170)]
19	0.002493	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=36/2016, ttl=64 (request id 380)]
20	0.002545	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=36/2016, ttl=64 (reply id 380)]
21	0.002750	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=40/3072, ttl=64 (request id 200)]
22	0.002802	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=40/3072, ttl=64 (reply id 200)]
23	0.003007	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=44/3520, ttl=64 (request id 250)]
24	0.003059	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=44/3520, ttl=64 (reply id 250)]
25	0.003264	192.168.0.101	192.168.0.181	ICMP	60	Echo (ping) request [seq=48/3096, ttl=64 (request id 237)]
26	0.003316	192.168.0.181	192.168.0.101	ICMP	60	Echo (ping) reply [seq=48/3096, ttl=64 (reply id 237)]

Packet 1: Packet, 60 bytes on wire (784 bits), 60 bytes captured (768 bits) on interface eth0, id 0
 Ethernet II, Src: hp-af:57:85 [192.168.0.101], Dst: DlinkSystem-63:bf:7f [192.168.0.181]
 Internet Protocol Version 4, Src: 192.168.0.101, Dst: 192.168.0.181
 Internet Control Message Protocol

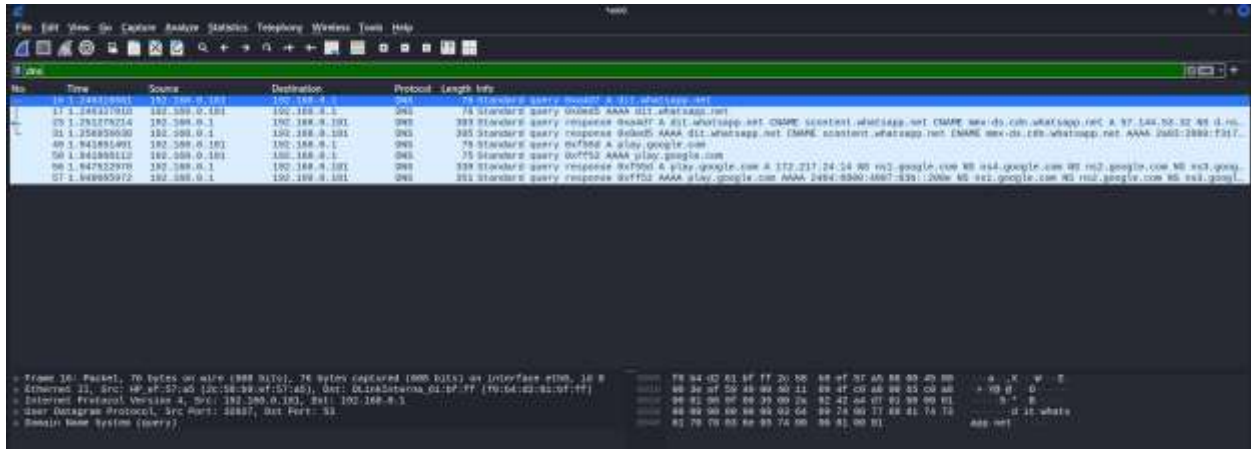
2. DNS (Domain Name System)

- Used to resolve domain names into IP addresses.

Observations:

- DNS queries sent from 192.168.0.181 to the local DNS server 192.168.0.1
- Domain names resolved include:
 - google.com
 - play.google.com

- dit.whatsapp.net
- Both **A (IPv4)** and **AAAA (IPv6)** record queries were observed
- DNS responses contained multiple IP addresses and CNAME records



3. TCP (Transmission Control Protocol)

- Provides reliable, connection-oriented communication.

Observations:

- TCP connections observed after DNS resolution
- Three-way handshake detected:
 - SYN
 - SYN-ACK
 - ACK
- Used for application traffic such as HTTPS and other services
- Ensures reliable delivery and retransmission if packets are lost

No.	Time	Source	Destination	Protocol	Length	Info
302	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
303	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
304	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
305	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
306	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
307	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
308	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
309	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
310	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
311	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
312	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
313	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
314	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
315	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
316	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
317	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
318	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
319	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
320	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
321	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
322	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
323	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
324	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
325	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0
326	14.642433000	192.168.0.181	192.168.0.181	TCP	60	5580 → 5580 [ACK] Seq=14451352 Win=0 Len=0

4. Packet Details Summary

Field	Example Value
Source IP	192.168.0.181
Destination IP	142.250.77.142
Protocol	ICMP
Packet Size	98 bytes
TTL	64 (request), 119 (reply)
DNS Server	192.168.0.1
DNS Port	UDP 53

6. Conclusion

This experiment successfully demonstrated how live network traffic can be captured and analyzed using Wireshark. Multiple protocols such as ICMP, DNS, and UDP were identified, showing how different layers of network communication work together. ICMP packets confirmed network connectivity, while DNS traffic revealed domain name resolution processes.

This exercise provided hands-on experience in packet analysis, protocol identification, and basic network troubleshooting, which are essential skills in cybersecurity.

7. Tools Used

- Wireshark
- Linux Terminal
- Ping utility